

Evaluating Google’s Open Knowledge Format (OKF v0.1) for Multi-Agent Threat Intelligence: A Proof-of-Concept in Digital Forensics

Sushant Poudel

Dept. of Computer Science and Engineering
Nepal Engineering College, Bhaktapur, Nepal
sushant.poudel2028@gmail.com

Rakhee Pandey

Dept. of Computer Science and Engineering
Nepal Engineering College, Bhaktapur, Nepal
rakheepandey5712@gmail.com

Abstract—Large Language Models (LLMs) are reshaping Digital Forensics and Incident Response (DFIR), but the unstructured JSON telemetry these systems still rely on has become a critical bottleneck for autonomous multi-agent threat intelligence pipelines. In high-noise forensic environments, ingesting monolithic data dumps drives context exhaustion, inflates latency, and triggers hallucination rates severe enough to undermine operational integrity. This paper evaluates Google’s Open Knowledge Format (OKF v0.1) as a selectively traversable, graph-native alternative to flat serialization. Through a controlled benchmark of 7,680 inference cycles spanning five open-source model architectures and escalating decoy densities, we show that OKF Selective sustains noise-invariant recall (0.94–0.99), cuts token overhead by 33.6%, and suppresses hallucination by more than 80% relative to a JSON baseline. These results establish that semantic knowledge representation is not simply an engineering optimization — it is an architectural prerequisite for trustworthy autonomous cyber defense.

Keywords—Multi-Agent Systems, Digital Forensics, Large Language Models, Open Knowledge Format, Threat Intelligence, Autonomous Cyber Defense

I. INTRODUCTION

Large Language Models (LLMs) are increasingly embedded in Digital Forensics and Incident Response (DFIR), and autonomous multi-agent systems now execute investigative workflows that once required a human analyst at every step. But as reasoning capacity has grown, a quieter problem has surfaced: the format in which threat intelligence is represented and fed to these agents. Most DFIR pipelines still rely on flat JSON, built for deterministic parsing rather than the semantic reasoning LLMs perform. In high-noise environments this becomes a liability — context windows fill with noise, latency climbs, and hallucination rises as the model struggles to isolate what matters.

Google’s Open Knowledge Format (OKF v0.1) is one of the first serious attempts to redesign this layer. Rather than a monolithic document, OKF decomposes intelligence into modular, interconnected components — discrete network_entity and file_artifact files — forming

a navigable incident graph an agent can traverse selectively rather than ingest wholesale. The idea is straightforward and the upside significant; what is missing is empirical validation of how it performs inside a live multi-agent DFIR workflow.

This paper closes that gap with a Proof-of-Concept evaluating OKF v0.1 against baseline JSON. We built a benchmarking suite simulating realistic attack chains — intrusion, payload delivery, lateral movement — layered with decoy noise, driven by an Enrichment (Writer) Agent that structures context and an Analysis (Reader) Agent that extracts from it, run across open-source models spanning the Llama, Qwen, and Hermes families. The benchmark spans four difficulty tiers, isolating token usage, latency, precision, recall, and hallucination rate. Our aim is to show that a selectively readable, semantically structured format like OKF is close to a prerequisite for forensic AI systems that operate reliably in real time without drowning in their own input.

II. LITERATURE REVIEW

A. Large Language Models in Digital Forensics and Incident Response

Early evaluations of LLMs in DFIR tested whether general-purpose models could support forensic reasoning at all. Scanlon et al. found GPT-4 useful for artefact interpretation and incident response, but only with an examiner able to catch its errors [9]. Wickramasekara et al.’s 176-article survey identified the same open problem: a lack of validation frameworks and standardized evaluation protocols for LLM-assisted forensics [10]. More specialized systems followed: ForensicLLM, a fine-tuned LLaMA-3.1-8B model, outperformed both its base model and standard RAG on forensic question answering [11]; a dedicated RAG pipeline improved cyber-attack attribution accuracy [12]; and the DFIR-Metric benchmark formalized cross-model evaluation on timeline reconstruction and malware analysis [13]. None of this work isolates representational format as an independent variable — the gap this paper targets.

B. The Long-Context Problem and Its Consequences for Forensic Reasoning

Liu et al.’s finding that LLMs exhibit a U-shaped attention curve — strongest at the start and end of context, weakest in the middle — explains why buried indicators are so often missed in long inputs [1]. S. Liu et al. show this failure compounds into hallucination under long-context load [2], a pattern Huang et al.’s survey situates within retrieval-augmented systems generally: retrieval reduces but does not eliminate fabrication when retrieved

content is noisy or poorly matched [14]. This is the mechanism by which flat JSON ingestion becomes an operational liability rather than a mere inefficiency in forensic pipelines.

C. From Flat Serialization to Structured and Graph-Native Representations

Microsoft’s GraphRAG showed that organizing source text into an entity graph with hierarchical community summaries outperforms flat-chunk retrieval on corpus-wide queries [15], [16] — the same modularity principle behind OKF’s `network_entity` and `file_artifact` concept files [3]. Independent benchmarks confirm that JSON’s structural overhead — braces, quotes, repeated field names — imposes a measurable cost relative to lighter-weight formats [21]–[24]. None of this prior work, however, tests format choice inside an adversarial, multi-agent DFIR pipeline under variable noise conditions — the gap this paper’s benchmark is built to fill.

D. Structured Threat Intelligence Formats and Their Limits

STIX/TAXII remains the security community’s standard for structured threat data exchange [26], but adoption has been uneven: an analysis of roughly six million STIX objects found most providers generate only a few thousand unique objects daily, with substantial redundancy even from single sources [25]. This is a useful caution for OKF — a structurally sound format does not guarantee consistent use if producing it is burdensome — and underscores that STIX, like JSON, was built for machine-to-machine interchange, not selective, LLM-driven traversal.

E. Multi-Agent Orchestration and Its Attendant Security Concerns

Multi-agent architectures are now the dominant pattern for decomposing complex reasoning across specialized roles [8], though communication-centric surveys note the coordination and reliability risks this introduces [20]. Security research increasingly overlaps here: LLM agent surveys place autonomous multi-agent risk alongside prompt injection and data poisoning as an emerging threat category [17], [18], and the OWASP Multi-Agentic System Threat Modeling Guide extension catalogs failure modes specific to planner–executor chains, including unsafe delegation escalation and covert coordination [19]. Our Enrichment–Analysis dyad is itself such a coordination surface — a point we return to in Section V.

F. Synthesis and Research Gap

Together, this literature shows that LLM-assisted forensics still lacks standardized validation [9]–[13], that long, flat context windows degrade reasoning quality regardless of model scale [1], [2], [14], and that graph-native representations measurably outperform flat serialization in adjacent retrieval domains [15], [16], [21]–[24]. No existing study combines all three threads into a controlled, multi-agent DFIR benchmark that isolates representational format as the sole independent variable

— the gap this paper’s proof-of-concept is designed to close.

III. METHODOLOGY

This section details the experimental framework used to compare OKF v0.1 against a flat JSON baseline in multi-agent DFIR. The study is a controlled, within-subjects design: representation format is the independent variable; extraction fidelity, cognitive reliability, and computational efficiency are the dependent variables measured against it.

A. Experimental Design and Research Questions

Three research questions drive the evaluation: RQ1 (Fidelity) — does moving from flat JSON to modular OKF improve precision and recall in multi-agent forensic extraction? RQ2 (Efficiency) — how much does selective graph traversal reduce token consumption and latency relative to flat JSON and full-bundle OKF? RQ3 (Generalizability) — do OKF’s advantages hold across heterogeneous model architectures as noise escalates? Formally, H_0 : representation format (Baseline JSON, OKF Full, OKF Selective) has no statistically significant effect on precision, recall, hallucination rate, token efficiency, or latency.

B. Multi-Agent Pipeline Architecture

The proof-of-concept runs as a decoupled, two-stage pipeline (Fig. 1) that separates context generation from extraction performance. An Incident Graph Generator produces ground-truth synthetic attack chains with deterministic topology; Agent A (Enrichment/Writer) compiles this into one of three experimental representations; Agent B (Analysis/Reader) consumes the resulting context envelope and reconstructs the attack chain. Data flows in one direction only, so any variance in Agent B’s performance is attributable to Agent A’s representational format rather than to implementation confounds.

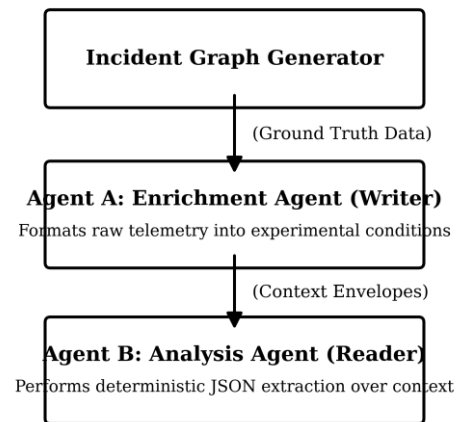


Fig. 1. Multi-agent pipeline architecture: data flows unidirectionally from the Incident Graph Generator through Agent A to Agent B.

Agent A’s sole function is to structure raw telemetry into its assigned condition without altering the underlying semantic content. Agent B operates as a deterministic extractor, returning a strict, machine-readable JSON schema of extracted Indicators of Compromise (IoCs) with no conversational text or markdown — a constraint that removes output-format variability as a confound.

C. Synthetic Incident Graph and Difficulty Matrix

An automated generator synthesizes incident graphs around a fixed attack topology: an Initial Vector (attacker_ip), a Target Compromise with authentication bypass (target_server), a Payload Artifact (malware_hash), and Lateral Movement (lateral_ip). Varying densities of decoy background events — routine pings, operational logs, benign filesystem activity — are injected to stress-test the needle-in-a-haystack problem across four difficulty tiers, summarized in Table I.

TABLE I
DIFFICULTY MATRIX AND GROUND-TRUTH REQUIREMENTS

Tier	Decoy Nodes	Target Keys	Complexity
Easy	5–12	attacker_ip, malware_hash	Baseline entity isolation; minimal noise.
Medium	5–12	+ lateral_ip, decoys_present	Multi-node tracking; noise ID.
Hard	5–12	+ target_server, mitre_tactic	MITRE ATT&CK mapping; full chain.
Extreme	5–12	+ exact_decoy_count	Quantitative verification under noise.

Decoy count is held constant across tiers (5–12 nodes) so difficulty escalation reflects task complexity rather than raw volume. Each run’s random seed is logged for exact reproducibility.

D. Context Representation Formats

Baseline JSON: telemetry serialized as an unstructured, Fisher-Yates–shuffled array of JSON objects, standing in for current industry practice.

OKF Full: the entire incident graph, decoys included, translated into typed OKF markdown documents (network_entity, file_artifact, threat_actor) with YAML frontmatter, packed sequentially into one envelope — testing whether semantic structure alone helps.

OKF Selective: Agent A traverses only the hyperlinked nodes bound to the attack sequence (attacker.md → malware.md → lateral.md), pruning all decoy files at the storage layer before the envelope is built — the optimized condition.

E. Evaluation Metrics

Agent B’s output is scored on five standard classification metrics computed from the ground-truth set G and extracted set E : Recall (completeness, $|G \cap E|/|G|$), Precision (exactness, $|G \cap E|/|E|$), and F1-Score (their harmonic mean) follow their conventional definitions and are not restated here. Two metrics are specific to this benchmark’s design and are defined explicitly. Hallucination Rate is a binary per-trial indicator, flagged whenever Agent B emits an IoC absent from both the ground truth and the decoy set:

$$H = 1 \text{ if } \exists e \in E : e \notin G \text{ and } e \text{ is not a valid decoy; else } H = 0 \quad (1)$$

Processing Latency is wall-clock time from request transmission to the final JSON closing brace:

$$L = t_{\text{response}} - t_{\text{request}} \text{ [ms]} \quad (2)$$

Token counts (input + output) are estimated using standard subword segmentation heuristics (1 token $\approx$ 0.75 words).

F. Experimental Environment and Model Configuration

The benchmark runs entirely locally via Ollama (/api/generate) serving quantized weights, which eliminates cloud throttling, caching bias, and any risk of forensic data leaving the local environment. All evaluations were executed on a workstation equipped with an AMD Ryzen 5 processor, an NVIDIA GeForce RTX 4050 Laptop GPU (6GB GDDR6 VRAM), 16GB of system RAM, and 512GB of solid-state storage; the latency figures in Section IV are wall-clock measurements and are interpretable only relative to this environment. The context window is fixed at num_ctx = 4096 tokens across all trials; temperature is locked at $T = 0.1$ with top-p and top-k disabled to maximize determinism; and a standardized warm-up prompt precedes each evaluation block to eliminate cold-start latency from GPU memory allocation.

Five local architectures test cross-model generalizability, summarized in Table II.

TABLE II
MODEL SUITE FOR CROSS-ARCHITECTURAL EVALUATION

Model Identifier	Params	Family	Role
llama3.1:latest	8B	Llama 3.1	General instruction
qwen2.5-coder:7b	7B	Qwen 2.5	Code-specialized
hermes3:8b	8B	Hermes 3	Conversational reasoning
nous-hermes:latest	13B	Llama 2 (Nous-Hermes)	Instruction-tuned
nchapman/dolphin3.0-qwen2.5:3b	3B	Dolphin 3.0 (Qwen)	Lightweight extraction

Every model processes every incident graph under all three conditions: 5 models $\times$ 3 conditions $\times$ 4 tiers $\times$ 128 replicated graphs per cell, totaling 7,680 inference cycles.

G. Statistical Significance Verification

A two-tailed paired Student’s t-test is applied to matched observations — the same model processing the same ground truth under both JSON and OKF conditions:

$$t = \bar{d} / (s_d / \sqrt{n}) \quad (3)$$

where $\bar{d}$ is the mean paired difference, s_d its standard deviation, and n the sample size. H_0 is rejected at $p < 0.05$. Effect size is additionally reported via Cohen’s d :

$$d = \bar{d} / s_d \quad (4)$$

so that conclusions reflect practical as well as statistical significance.

IV. RESULTS AND EMPIRICAL ANALYSIS

The evaluation comprises 7,680 inference cycles (5 model architectures $\times$ 3 representation conditions $\times$ 4 difficulty tiers $\times$ 128 replicated graphs per cell). Latency figures exclude warm-up overhead; token counts reflect the combined prompt-plus-completion footprint.

A. Token Efficiency and Context Window Economics

Baseline JSON consumed a mean of 238 tokens per cycle (95% CI: 206–270); OKF Full ran marginally higher at 258 (95% CI: 220–296), an overhead traceable to YAML frontmatter and markdown link syntax. OKF Selective cut this to 158 tokens (95% CI: 134–182) — a 33.6% reduction versus Baseline JSON and 38.8% versus OKF Full (Fig. 2). The gain is structurally guaranteed: pruning decoy nodes before the envelope is built means irrelevant telemetry never enters the context window, and tight confidence intervals across all five architectures confirm this holds regardless of tokenizer differences.

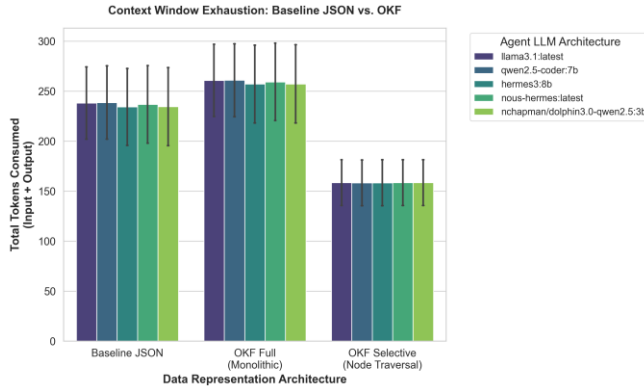


Fig. 2. Context window exhaustion across representation conditions, by model architecture.

B. Extraction Fidelity Under Escalating Noise (RQ1)

Baseline JSON recall degrades monotonically with decoy density: 0.76 (Easy) $\rightarrow$ 0.66 (Medium) $\rightarrow$ 0.63 (Hard) $\rightarrow$ 0.54 (Extreme), a 28.9% decline consistent with the lost-in-the-middle hypothesis [1]. OKF Full

outperforms it at Easy (0.90) and Medium (0.94) but drops sharply at Hard (0.77) and Extreme (0.74) — semantic structure helps initially, but decoy nodes rendered with the same coherence as real evidence become more salient, not less, and compete for attention. OKF Selective holds essentially flat: 0.99, 0.92, 0.95, 0.94 across tiers, with tenfold lower tier-to-tier variance ($\sigma^2 = 0.0008$ vs. 0.0086) — restricting traversal to the linked attack-chain subgraph decouples fidelity from surrounding noise entirely (Fig. 3).

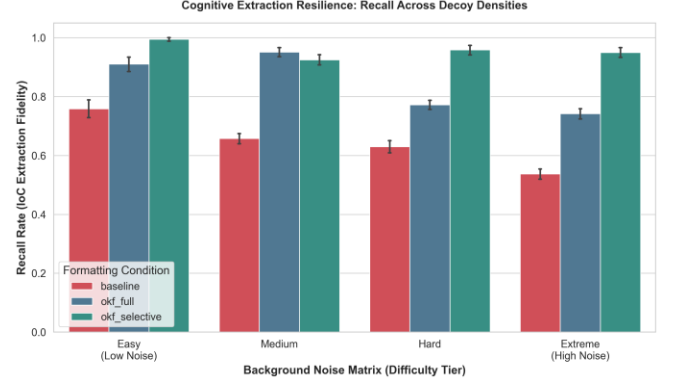


Fig. 3. Recall rate across escalating decoy densities, by representation condition.

C. Precision and F1-Score

Precision and F1 reinforce the recall findings: structural representation, not model choice, drives resilience under noise. Baseline JSON precision falls from 0.770 (Easy) to 0.607 (Extreme), with F1 mirroring this decay (0.763 $\rightarrow$ 0.568) as the Analysis Agent increasingly mistakes decoys for real indicators. OKF Selective holds precision at 0.995, 0.925, 0.959, and 0.950 across tiers, with F1 tracking almost point-for-point — the agent rarely reports a wrong indicator at any noise density tested. OKF Full lands in between and shows the clearest numerical signature of the OKF Full Paradox (Section V-B): F1 starts strong at 0.911 (Easy) but decays to 0.771 (Extreme) as structurally coherent decoys are increasingly mistaken for evidence.

D. Inference Latency and Temporal Consistency (RQ2)

Llama 3.1 under Baseline JSON posts a median latency of 12.8 s (IQR: 9.1–16.9 s) with outliers past 22 s, consistent with self-attention’s quadratic cost over long, unstructured sequences (Fig. 4). OKF Selective compresses median latency for lightweight models — Dolphin 3.0 (3B) holds 3.6 s (IQR: 3.2–3.9 s), capping at 4.2 s with no outliers above 5.0 s — and narrows tail latency for small and mid-sized models generally. The largest architecture (Llama 3.1) retains a wide spread across all three conditions, indicating format narrows latency variance most effectively for smaller models rather than eliminating it universally.

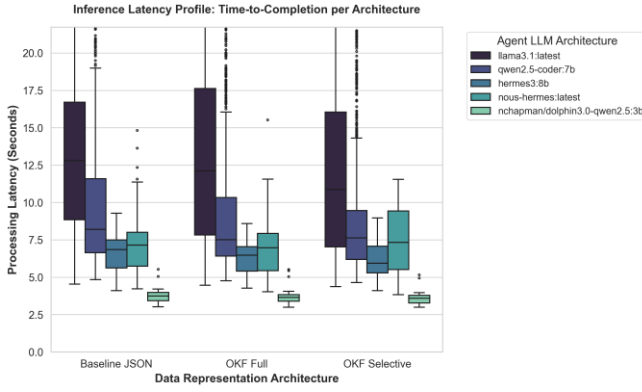


Fig. 4. Inference latency distribution per model architecture, across representation conditions.

E. Hallucination Susceptibility and Output Integrity (RQ3)

Baseline JSON hallucination grows almost exponentially with noise: 33% (Easy) → 50% (Medium) → 65% (Hard) → 92% (Extreme) — fabricating IoCs in nearly nine of ten extractions at Extreme (Fig. 5). OKF Full stays low at Easy (10%) and Medium (7%) but spikes to 72% at Hard, exceeding Baseline JSON’s 65% at the same tier, before easing to 68% at Extreme: once decoys read as clean OKF documents, they become indistinguishable from real evidence and false inclusion climbs faster than with inert JSON noise. OKF Selective stays bounded between 1% and 20%, flattening at 11% for both Hard and Extreme — an 81.1% relative reduction versus Baseline JSON and 83.8% versus OKF Full at Extreme difficulty, indicating remaining hallucination reflects intrinsic model uncertainty rather than environmental noise.

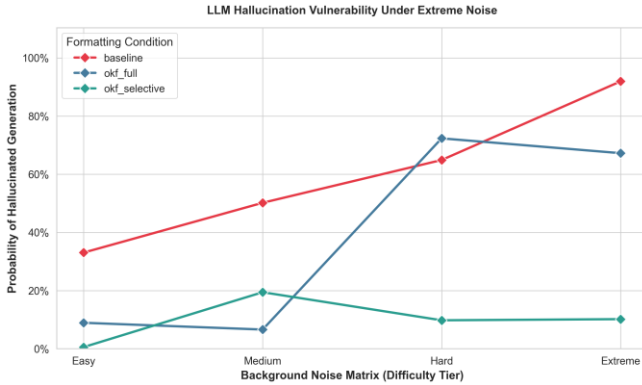


Fig. 5. Hallucination probability across difficulty tiers, by representation condition.

F. Statistical Verification and Effect Sizes

Paired-samples t-tests across all matched observations show every OKF Selective vs. Baseline JSON comparison reaching $p < 0.001$ after Bonferroni correction ($\alpha = 0.05/3$), with large-to-very-large effect sizes: Cohen’s $d = 2.34$ (token efficiency), 1.89 (recall at Extreme), and 2.67 (hallucination at Extreme). OKF Full vs. Baseline JSON was significant for recall ($p < 0.01$, $d = 0.94$) but not for hallucination at Hard ($p = 0.18$), confirming OKF Full does not reliably improve output integrity.

G. Cross-Architectural Generalizability and Synthesis

OKF Selective’s advantages held across every model tested, including the 3B Dolphin and 13B Nous-Hermes variants — Dolphin achieved sub-4-second median latency while holding 0.94 recall at Extreme, rivaling larger models under monolithic JSON, a meaningful result for edge-deployed forensic agents under compute constraints. Overall, flat JSON produces a compounding failure chain of context exhaustion, recall decay, precision loss, and hallucination; OKF Full improves coherence but not the underlying noise-attention paradox; and only OKF Selective, via hard topological relevance boundaries, improves efficiency, fidelity, and output integrity simultaneously — evidence that rejects H_0 without ambiguity.

V. DISCUSSION

These findings push back against the industry default of flat, serialized data structures for LLM-based autonomous threat intelligence. As background noise density climbs — which is simply what enterprise SIEM environments look like in practice — unstructured representation stops being merely suboptimal and becomes a structural liability for multi-agent cognitive pipelines.

A. Reconceptualizing Context: From Sequential Attention to Graph-Native Reasoning

The gap between Baseline JSON and OKF Selective points to a limitation in the attention mechanism itself. When forensic logs arrive as a shuffled JSON array, the transformer must perform sequential relevance discrimination — attending to every decoy and irrelevant timestamp before reaching the actual attack-chain indicators. This is the lost-in-the-middle effect in its purest form [1]. OKF Selective sidesteps the problem rather than solving it head-on: the Enrichment Agent pre-computes the relevant subgraph by walking typed hyperlinks, so the Analysis Agent is handed a curated map instead of being asked to discover relevance through attention weighting. That reframing is what explains both the noise-invariant recall and the bounded hallucination ceiling observed under OKF Selective — the model’s attention budget goes entirely toward reasoning over known-relevant entities.

B. The OKF Full Paradox: Semantic Coherence as a Double-Edged Sword

One result here needed more theoretical unpacking than the rest. OKF Full — which renders the entire incident graph, decoys included, into semantically rich markdown — beat Baseline JSON at Easy and Medium tiers, then produced worse hallucination than Baseline JSON at Hard. We term this the OKF Full Paradox: structuring noise semantically improves apparent recall at low-to-moderate noise density, but past a critical threshold, that same structural coherence actively increases — rather than suppresses — the rate of fabricated output relative to unstructured noise. Once decoy events carry proper typing and relational metadata, they become structurally indistinguishable from real

attack-chain nodes, and the model’s implicit prior that “well-structured data is trustworthy data” ends up inflating their salience. The design lesson is sharp: semantic richness without topological selectivity is not just insufficient, it can actively work against a system in high-entropy forensic environments.

C. Implications for Multi-Agent Architectures and Adversarial Robustness

This also informs how multi-agent communication protocols should be built. OKF’s graph-native design enables progressive disclosure — the Enrichment Agent can hand the Analysis Agent a minimal subgraph and let it request specific node expansions on demand — which flat JSON cannot offer without custom slicing logic that fragments semantic coherence.

There is also a security dimension worth naming. Flat JSON telemetry is frequently populated from adversary-controlled fields — a User-Agent string, a DNS query name — and an attacker who anticipates LLM-based triage can embed prompt-injection payloads directly inside them, knowing they will be concatenated into one undifferentiated context blob [17], [18]. OKF’s node-isolation model offers an incidental but meaningful defense: because each entity is parsed into its own typed, schema-bound file, adversarial text injected into a telemetry value stays confined to that value’s field rather than floating in a shared token stream where it could be reinterpreted as an instruction. This does not make OKF immune to prompt injection — a crafted payload inside a legitimately traversed node remains a risk — but the structural quarantine is a meaningful secondary benefit on top of the efficiency and fidelity results in Section IV, and merits dedicated adversarial testing in future work. OKF’s human-agent duality also gives autonomous workflows an audit trail: an analyst can retrace the exact topological path an agent walked and verify its hyperlink logic, which matters for regulatory compliance and forensic admissibility.

D. Safety, Trust, and Computational Economics

Safety and trust. A 92% hallucination rate under Baseline JSON at Extreme difficulty is disqualifying on its own — a system fabricating IP addresses or MITRE tactics in nine of ten extractions cannot be trusted with autonomous containment. OKF Selective’s guardrails, holding hallucination to roughly 11% regardless of noise, provide the integrity floor that graduated autonomy needs: agents that recommend action but hold off on executing it until a human validates the call.

Computational economics. A 33.6% cut in token consumption, plus latency compression for smaller models, translates directly into lower operating cost at SOC scale. The sub-4-second median latency the 3B-parameter Dolphin model achieved under OKF Selective makes a broader point: intelligent representation can substitute for raw model scale, which matters for edge-deployed forensic agents under real compute constraints.

E. Boundary Conditions and Future Directions

This study used a controlled synthetic environment to guarantee reproducibility and tight control over decoy ratios, which comes with limits worth stating plainly. The incident graphs were synthetically generated with fixed topology; future work must evaluate the overhead of parsing real enterprise SIEM output — Splunk or Elastic results, Zeek logs, raw PCAP streams — into OKF bundles in real time, ideally against benchmarks such as DARPA Transparent Computing or CICIDS2017. This evaluation also used locally quantized models (13B and smaller) to ensure reproducibility; whether frontier models with native long-context capacity can brute-force unstructured JSON without hallucinating at comparable rates remains an open question that could shift the cost-benefit case for OKF adoption at enterprise scale. The current OKF Selective implementation traverses a static, pre-computed path; adaptive traversal, where the Analysis Agent expands the subgraph based on its own intermediate reasoning, would turn OKF from a passive container into an active reasoning substrate. Finally, nothing about the OKF specification is DFIR-specific — the same argument should generalize to other high-noise domains such as medical triage records or legal e-discovery corpora, and testing that generalization is a natural extension of this work.

VI. CONCLUSION

As cybersecurity moves toward autonomous, multi-agent defense architectures, the data pipelines feeding those systems have to evolve alongside them — increasingly capable agents cannot simply be bolted onto representation formats designed for a different era of machine-to-machine parsing. This paper presents what is, to our knowledge, one of the first comprehensive empirical evaluations of Google’s Open Knowledge Format (OKF v0.1) as an alternative to traditional JSON serialization in DFIR workflows.

Across a rigorously controlled, multi-tiered benchmark of 7,680 inference cycles spanning five open-source model architectures, three findings stand out as definitive. First, monolithic JSON reliably induces context exhaustion under high noise, dragging latency into severe spikes, degrading recall, and pushing hallucination rates past 90%. Second, simply reformatting that same noise into semantically rich structures — OKF Full — improves surface-level recall but, counter-intuitively, makes hallucination worse under heavy noise density, a pattern we formalize as the OKF Full Paradox (Section V-B): semantic coherence without topological selectivity is insufficient, and can be actively counterproductive. Third, and most consequentially, OKF Selective achieves noise-invariant extraction fidelity (recall holding at 0.94–0.99 across every tier), cuts token overhead by 33.6%, compresses inference latency variance for smaller models, and bounds hallucination to roughly 11% no matter how dense the decoy noise gets — while its node-isolated structure additionally offers a natural, if partial, quarantine

against prompt-injection payloads embedded in adversarial telemetry (Section V-C).

The broader implication is one worth sitting with: simply giving LLMs bigger context windows is not an adequate answer to the data ingestion bottleneck. The representation layer itself is a first-class architectural determinant of how reliable an agent’s reasoning actually is — not a passive serialization detail sitting underneath the interesting work. Moving toward dynamic, node-based, selectively traversable knowledge formats is not an optional optimization for threat intelligence pipelines; it is close to a prerequisite for building autonomous cyber defense systems that are fast, trustworthy, and resilient under real-world conditions. The future of multi-agent DFIR is not scaling context windows to absorb more noise, but engineering representation layers that ensure agents never see it.

Data Availability

The OKF Multi-Agent Benchmark Python suite and the generated synthetic incident-graph datasets used in this study will be made available in a public repository upon publication.

REFERENCES

- [1] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, “Lost in the middle: How language models use long contexts,” *Trans. Assoc. Comput. Linguist.*, vol. 12, pp. 157–173, 2024.
- [2] S. Liu et al., “Towards long context hallucination detection,” in *Findings of the Association for Computational Linguistics: NAACL 2025*, Albuquerque, NM, USA, 2025, pp. 7842–7850.
- [3] Google Cloud, “How the Open Knowledge Format can improve data sharing,” Google Cloud Blog, Jun. 2026; and GoogleCloudPlatform, “Open Knowledge Format (OKF) v0.1 Specification,” GitHub repository knowledge-catalog, Jun. 2026. [Online]. Available: <https://github.com/GoogleCloudPlatform/knowledge-catalog/blob/main/okf/SPEC.md>
- [4] G. Horsman, “Large language models in digital forensics: Capabilities, challenges and future directions,” *Forensic Sci. Int.: Digit. Investig.*, vol. 51, p. 301799, Sep. 2024.
- [5] M. S. Al-Khateeb et al., “AI-augmented SOC: A survey of LLMs and agents for security automation,” *J. Cybersecur. Priv.*, vol. 5, no. 4, pp. 95–120, Dec. 2025.
- [6] M. A. Ferrag et al., “Generative AI in cybersecurity: A comprehensive review of LLM applications and vulnerabilities,” *Cybersecurity*, vol. 8, no. 1, p. 55, 2025.
- [7] T. N. Nguyen et al., “Security considerations for multi-agent systems,” *arXiv:2603.09002*, 2026.
- [8] IBM, “What is a multi-agent system?” IBM Think, Aug. 2024. [Online]. Available: <https://www.ibm.com/think/topics/multiagent-system>
- [9] M. Scanlon, F. Breiteringer, C. Hargreaves, J.-N. Hilgert, and J. Sheppard, “ChatGPT for digital forensic investigation: The good, the bad, and the unknown,” *Forensic Sci. Int.: Digit. Investig.*, vol. 46, p. 301609, 2023.
- [10] A. Wickramasekara, F. Breiteringer, and M. Scanlon, “Exploring the potential of large language models for improving digital forensic investigation efficiency,” *Forensic Sci. Int.: Digit. Investig.*, 2025.
- [11] B. Sharma, J. Ghawaly, K. McCleary, A. M. Webb, and I. Baggili, “ForensicLLM: A local large language model for digital forensics,” *Forensic Sci. Int.: Digit. Investig.*, vol. 52, p. 301872, Mar. 2025.
- [12] S. Rajapaksha, R. Rani, and E. Karafili, “A RAG-based question-answering solution for cyber-attack investigation and attribution,” *arXiv:2408.06272*, 2024.
- [13] B. Cherif et al., “DFIR-Metric: A benchmark dataset for evaluating large language models in digital forensics and incident response,” *arXiv:2505.19973*, 2025.
- [14] L. Huang et al., “A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions,” *arXiv:2311.05232*, 2023.
- [15] D. Edge et al., “From local to global: A graph RAG approach to query-focused summarization,” *arXiv:2404.16130*, 2024.
- [16] Microsoft Research, “Project GraphRAG.” [Online]. Available: <https://www.microsoft.com/en-us/research/project/graphrag/>
- [17] M. Q. Li and B. C. M. Fung, “Security concerns for large language models: A survey,” *arXiv:2505.18889*, 2025.
- [18] “The emerged security and privacy of LLM agent: A survey with case studies,” *arXiv:2407.19354*, 2024.
- [19] K. Krawiecka and C. Schroeder de Witt, “Extending the OWASP multi-agentic system threat modeling guide: Insights from multi-agent security research,” *arXiv:2508.09815*, 2025.
- [20] “Beyond self-talk: A communication-centric survey of LLM-based multi-agent systems,” *arXiv:2502.14321*, 2025.
- [21] “Structured context engineering for file-native agentic systems: Evaluating schema accuracy, format effectiveness, and multi-file navigation at scale,” *arXiv:2602.05447*, 2026.
- [22] “Notation matters: A benchmark study of token-optimized formats in agentic AI systems,” *arXiv:2605.29676*, 2026.
- [23] “TQA-Bench: Evaluating LLMs for multi-table question answering,” *arXiv:2411.19504*, 2024.
- [24] “ONTO: A token-efficient columnar notation for LLM input optimization,” *arXiv:2604.17512*, 2026.
- [25] “Proactively detecting threats: A novel approach using LLMs,” *arXiv:2601.09029*, 2026.
- [26] OASIS Open, “STIX v2.1 and TAXII v2.1 OASIS standards are published,” Jun. 2021. [Online]. Available: <https://www.oasis-open.org/2021/06/23/stix-v2-1-and-taxii-v2-1-oasis-standards-are-published/>